8 Here are the first five numbers in a sequence.

47 41 35 29 23

(a) Find the first negative number in the sequence.

Subtract6 each time: 23,17,11,5,-1.

1 mark
Continue subtracting6

First negative number = -1.

1 mark
−1

(2)

Sarah says,

" –100 is **not** a number in this sequence."

(b) Is Sarah correct?
Explain why.

1 mark
Odd terms,even−100

Yes. Every term is odd: subtracting6 preserves oddness.

-100 is even, so it cannot occur.

(1)

(Total for Question 8 is 3 marks)